NATHAZ AI LAB / TECHNICAL REPORT · EXPLORATORY TRIAL

L22.3 Uses of self-built / local models for our research

Group Idea Lab

Status Closed (negative result)

Period 2026-10-13 06:06 — 2026-10-24 21:54

Stage conclusion

Project report

Uses of self-built / local models for our research

Objective (redirected by the PI, 10-13 10:38)

Answer the PI's original question: what are the uses of **self-built / local large language models (LLMs)** for our research? Four questions: (1) Can local open-source LLMs take over batch routine tasks (literature screening, PDF data extraction, knowledge base tagging, identifying thunderstorm winds from Storm Events narratives, and diagnosing supercomputing job logs)? (2) Is fine-tuning (LoRA / full-parameter) on our materials worthwhile in a head-to-head comparison against "frontier model + retrieval"? (3) How are other domain LLMs (OceanGPT, ClimateGPT, GeoGPT, etc.) built, what are their costs and evaluations, do any exist in wind engineering / civil engineering, and would building one ourselves have publication value? (4) How do they differ from Claude/Gemini/Codex/Grok on our real tasks? Task brief.

Current status

10-13 13:32: conclusion reached; status changed to On hold (questions answered; operational recommendations handed over to the platform rather than kept as a research project). The sub-project "Forecasting uses of AI weather foundation models" proceeds separately. **In one sentence**: Do not build our own large language model (continued pre-training / full fine-tuning); the first step is to offload batch routine tasks from Claude—hand initial literature abstract screening and single-value PDF extraction to the Gemini 3.8 Flash batch channel (best quality, already covered by subscription) or Qwen3-32B on Colab (when data should not be sent externally or offline processing is required), use keyword rules first wherever possible, and reserve Claude for judgment and review. **Restart conditions**: Conduct a LoRA comparison only when a large volume of fixed-format, labeled narrow tasks (such as internal configuration generation) arises and local models show a clear gap behind frontier models; enable the local channel when large batches of text that should not be transmitted externally appear.

Assessment table (large language models; after two evaluation rounds, 10-13 13:32)

Question	Our specific use case	Existing data	Minimal experiment	Cost	Recommendation
1 Local open-source LLMs taking over batch routine tasks	Initial literature abstract screening, single-value PDF extraction	Knowledge base cards, Zotero PDFs, public NOAA data	**Two rounds completed** (Round 2 blind run on 350 frozen items)	Round 2 ~1.97 CU; ~5.6 CU per 1,000 items (including amortized cold start)	**Abstract screening: usable**—Qwen3-32B achieved 82% accuracy, 98% recall, and 58% precision, comparable to Claude Sonnet (80.5%/100%/56%), while Gemini 3.8 Flash performed best (85.5%/98%/64%), all far outperforming keyword rules (67.5%); human review of excluded borderline samples is still required. **Single-value PDF extraction: usable for initial extraction**—local 92% vs. Claude/Gemini 96%; original pages must be verified. **Storm Events thunderstorm wind identification: use rules** (post-construction rule accuracy 100%; genuinely ambiguous samples not tested). Log error diagnosis not tested.

Question	Our specific use case	Existing data	Minimal experiment	Cost	Recommendation
2 Fine-tuning on our materials (LoRA)	Fixed-format narrow tasks	Requires building a separate labeled dataset	Not done—local models are already comparable to frontier models, and the gap is too small to justify the effort	—	Low (see restart conditions above)
3 Building "our own domain LLM" (continued pre-training)	"Wind Engineering GPT"	Insufficient corpus volume	Do not do	Hundreds to thousands of A100·h minimum	Low (see review verdict for literature basis; civil engineering has the CivilGPT precedent, none found in wind engineering)
4 Gap vs. frontier models	Same as 1	Same as 1	Same as 1	—	Screening: local 32B ≈ Claude Sonnet < Gemini 3.8 Flash; data extraction: local is ~4 percentage points lower (2 items difference out of 50); Codex/Grok not tested

Question	Our specific use case	Existing data	Minimal experiment	Cost	Recommendation
Estimated usage savings (based on this line's data): Each batch call to Claude via CLI consumes ~39k tokens (mostly framework overhead), at a list price of ~$0.05–$0.06 per item; local Colab costs ~5.6 CU per 1,000 items (including amortized cold start). If there are 10,000 screening/extraction routine items per month, ~10,000 Claude calls can be offloaded entirely (to the Gemini subscription channel or locally). The proportion of total platform Claude usage this represents depends on the actual volume of such routine tasks—the platform does not track this, so a percentage cannot be given.					

Finalized decisions

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- Literature review verdict
- Physical-field surrogate section: (archived)

Trials and results (large language models)

Date	What was done	One-sentence result
10-13 01:15	Strata + Qwen3.8-Flash-Next (Colab A100, see)	94 GB-class model ran successfully at 32K context with 35–59 tok/s output, requiring ~40 min / 4 CU setup per new session; single-turn interaction is slow, batch throughput not tested
10-13 11:10	Review (3/3 conditional support) verdict: restore qualified positive comparisons for LLaMat/NL2FOAM, add CivilGPT and PMC12657656, retract unified MCP figures, and split ratings by task	
10-13 12:18	Round 1 evaluation on 100 items (Colab A100 40 GB, ~55 min ≈ 4.85 CU; Gemini CLI authentication failed, not compared)	A Screening: 8B scored 40/40, but keyword rules also scored 40/40 (items too easy); B PDF extraction: Qwen3-30B 24/25 = Claude 24/25; C Storm Events: best local 12/20 < Claude 15/20; D testing ratios alone was uninformative. Local run took 3–4 min for 100 items (including loading); Claude CLI used ~20k–40k tokens per item (including system overhead). Deviation: evaluation set was not submitted prior to running, and some items were re-tested after correction
10-13 13:17	Round 2 on 350 items (blind run after hash freeze; A100 80 GB, ~17 min ≈ 1.97 CU)	A2: local 32B 82% ≈ Claude 80.5%, Gemini 85.5%, keyword rules 67.5%; B2: local 92% vs. frontier 96%; C2: keyword rules 100% (items still too easy)
10-13 10:45	Literature review: --max returned (16 min, 34k characters) and verified by the lead; default tier returned (19 min, 24k characters) and verified; both submitted for review	Max tier contained misattributions (OpenFOAMGPT/NL2FOAM) and unverified comparisons (LLaMat vs. GPT-4o), which were removed
10-13 10:50	GitHub repository table (queried directly via the API by the lead)	Inference/fine-tuning/PDF parsing frameworks are mature and active (vLLM 94k★, LlamaFactory 75k★, unsloth 78k★, Apache/MIT); domain LLM repositories have few stars and are mostly inactive

Open items and next steps

- Answered in full. The Platform Lead endorsed the recommendation and forwarded it to the PI to decide the platform-wide policy (recommending a one-week trial on the Lab line + LitWiki first, with 10%

sampled for review); within this sub-project we have already implemented this policy (by default, with local models used only for data that should not be sent externally). The root lead decided on 10-13 to run an in-track trial (10-13 to 10-20, sampling 10% for Claude review, reverting if agreement <90%), forwarded. If the PI wants an evaluation on genuinely ambiguous Storm Events samples or log error diagnosis, an additional round can be run (≤10 CU).

- The two follow-up items for the digital twin wind tunnel (v8 rerun and W3 cross-prediction) belong to the archived surrogate section; when notified by their side, they will be forwarded to the root lead for separate arrangement and will no longer be handled under this sub-project.

Literature basis

- To be supplemented after the literature review. Currently available: VNP48R7H (NL2FOAM, Qwen2.5-7B LoRA), FV7BN78P, 5ZE3U2A8; FoamGPT (ML4PS 2025 Table 1).

Byproduct: physical-field surrogates (archived, on hold)

Status: On hold (decided by the PI on 10-13 10:38, no further investment; open a separate sub-project if pursued in the future). One-sentence conclusion: On our existing data, learned models did not outperform appropriate classical baselines—in leave-one-building-out evaluation across 13 TPU rectangular high-rise geometries, ML won 0/5 strictly interior buildings against geometric interpolation; for Silsoe LBM inflow → windward Cp(t), CNN tied with a linear transfer function; literature precedents (DeepUFNet, ScaleSplit-NO) do not conflict with this (different tasks and comparison baselines). Side findings: the PI's March–May legacy project graph-deeponet-wind (UrbanTALES, holdout R^2 ~0.79, no non-learning baseline found); Silsoe inflow predictability decreases monotonically with distance (decay length roughly estimated at ~30 m). Restart conditions: Open a separate sub-project when a need arises for reconstruction or correction given existing sensor taps / low-fidelity information, or when adding a non-learning baseline to graph-deeponet-wind. Reviews: , .

Current status

10-13 06:40: Assessment table revised per review verdict (3/3 conditional support); minimal experiment dispatched to ideas (: leave-one-building-out on TPU isolated high-rise database, ML vs. interpolation/Kriging, ≤10 CU). Deep Research failed to start due to Gemini API monthly spend cap (429) and was reported to the platform (platform confirmed it was the billing cap on the PI's account, now restored and rerun); precedents initially checked via knowledge base cards + quick web search. Data inquiries to other lines in progress: lbm , cfd , wrf , cm1 , windtunnel (replied), events.

Assessment table (after DR rerun + review verdict, 10-13)

Category	Our specific use case	Is existing data sufficient?	Minimal experiment	Estimated cost	Recommendation
1 Domain fine-tuned LLM / LoRA	Code/standard Q&A, CFD configuration generation	Requires curating training/evaluation sets	Do not do	0	Low: No evidence shows local fine-tuning outperforms a frontier model + retrieval + error-correction pipeline (FoamGPT ML4PS 2025 Table 1: Qwen3-8B 0→26.36%, GPT-4.1-mini 35.45→39.09% with error correction disabled; cards FV7BN78P, 5ZE3U2A8)
2a Wind pressure surrogate: isolated rectangular high-rise, mean/rms Cp, geometric interpolation	Rapid Cp estimation	Public TPU database is sufficient	**Done**: Leave-one-building-out, ML won 0/5 strictly interior buildings	≈0.06 CU	Low (limited to this scenario: fixed GBM/MLP under this protocol; peak values, interference, and non-rectangular shapes not tested)
2a′-light Database boundary extrapolation, sparse tap reconstruction, multi-fidelity correction	Fewer wind tunnel tests / fewer LES runs	TPU is public; multi-fidelity lacks paired LF/HF data (pending CFD/LBM replies)	Candidate: systematic extrapolation on TPU boundary geometries; or sparse tap reconstruction	1–3 CU	Conditional candidate (LBM inflow → wind pressure already tested: linear FIR ≥ CNN, so this variant is not supported): precedents exist (NF6S3H9N single building across wind directions; DeepUFNet sparse taps → SR 4/3.75; TPU T111), adaptation to our data/compute unverified

Category	Our specific use case	Is existing data sufficient?	Minimal experiment	Estimated cost	Recommendation
2a′-heavy Urban wind field surrogate	Rapid urban wind field estimation	The PI's legacy project graph-deeponet-wind already built a 2D pedestrian-level wind speed surrogate on 512 UrbanTALES cases (holdout R^2 ~0.79, not compared against non-learning baselines); 3D not pursued	The Lab will not duplicate; candidate: add a non-learning baseline comparison for the best model from the legacy project	Far exceeds 10 CU	Deferred due to compute constraints (ScaleSplit-NO full training takes 9.7–25.5 h, verified by reviewer A)
2b Off-the-shelf AI weather models coupled to WRF/CM1	Low-cost ensembles driving WRF, environmental screening	Public ERA5/GFS	Not scheduled	—	Low–Medium (0.25° global models show skill on environmental parameters but are smooth, Feldmann 2024; Pangu 1-day lead time, Hua 2025 GRL; our focus is mainly hindcasting; WRF line reply: a 0.01 K perturbation causes a ~5× difference in severe wind area, information that AI initial conditions/ensembles cannot provide —not recommended)
3 Specialized small models	Thunderstorm vs. non-thunderstorm wind classification, wind pressure channel anomaly detection, job log error diagnosis	Public ASOS 1-min data; wind tunnel line has no measured data, but CFD line has CAARC wind tunnel measurements across 351 channels at 500 Hz	Candidate: ASOS 1D-CNN vs. rule baseline	1–2 CU	Medium, conditional on whether the events line has a classification bottleneck (2T26I5GW p.12, ASOS internal 5-fold acc 86.7%, card pending review)

DR precedents vs. TPU experiment: no conflict (confirmed by review)

- The two items in DR that looked most like "conflicts" have been verified to exist: DeepUFNet (arXiv 2508.03183 v2 abstract: trained at SR=1.5, test generalization at SR=4 and 3.75 described as "satisfactory") and ScaleSplit-NO (Qin et al. arXiv 2609.38977: Montreal urban wind field one-step NMSE 65.8% lower than baseline).
- The reason there is no conflict is that **the tasks and comparison baselines differ**, not that the precedents are unreliable: 1. Task: DeepUFNet reconstructs/advances the full field given sparse sensor time histories on the target cylinder; ScaleSplit-NO performs temporal advancement of the flow field over the same urban district. The TPU experiment predicts Cp on an unseen building **given only geometric parameters and zero sensor taps**. 2. Baseline: The precedents compare against other neural networks (U-Net, etc.); the TPU experiment compares against geometric interpolation/Kriging. No verified precedent beats interpolation under "whole unseen building geometry + head-to-head comparison against interpolation under identical conditions." 3. Consistent pattern: Urban FNO itself reports limitations across significantly different urban morphologies (arXiv 2501.05499); on TPU, ML likewise only holds an advantage in database boundary extrapolation (T111).
- Conclusion: What is worth testing is "reconstruction and correction when sensor taps / low-fidelity information are available" and "boundary extrapolation," not "pure geometric interpolation."

Finalized decisions

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- Review verdicts ,

Trials and results

Date	What was done	One-sentence result
10-13	Read 6 required cards + + quick web search	2a literature: high R^2 on random holdouts within the same database, poor rms on extrapolation; no strict "leave-one-unseen-building-out" precedent
10-13	(08:25 rerun, 11 min)	Report had no DOIs; after verification: DeepUFNet extrapolation across side ratios could not be verified, ScaleSplit-NO could not be found by name, and thunderstorm wind classification was missed; no verified precedent conflicting with the TPU conclusion was found. Verification and revision proposal submitted for review, verdict issued (DeepUFNet and ScaleSplit-NO updated to verified, 2a′ split into light/heavy)

Date	What was done	One-sentence result
10-13	Review verdict	3/3 conditional support; 7 wording revisions, TPU protocol supplemented with leave-one-building-out, point-level regression, peak removal, and pre-specified criteria
10-13	CM1 lead reply	No ML done yet. Available to borrow: 102 Moore soundings + 47 minute-by-minute near-surface fields (800×800, 5 channels, ~34 GB, local) with per-case summary statistics / interpretation labels, plus another 52 cases running. Preferred surrogate: sounding → outflow statistics (peak wind, area, duration, whether organized), sample size 47 → ~99
10-13	**Discovered the PI's March–May legacy project** on the cluster (tip from the CM1 lead, verified read-only by the lead)	UrbanTALES public LES urban wind field dataset of 512 cases (1 m), building height map → pedestrian-height wind speed field 640×640; auto_v2–v6 totaled ~250+ training runs across 40 architectures with AI-driven hyperparameter search; best holdout (47 cases) median R^2 ~0.78–0.79, beating the 200-epoch U-Net baseline (0.72) by ~0.07; **compared only against neural baselines, no non-learning baseline found**; no updates after 5-12, unrecorded in knowledge base; further local copies exist in the PI's earlier workspace
10-13	CFD lead reply	No ML done; small sample size (single event, single wind direction), suitable only as a validation set. Available to borrow: D420 near-surface wind scenarios/profiles, CAARC/Silsoe/Case011Up Cp, **CAARC wind tunnel measurements across 3 configurations (351 channels, 500 Hz, 120 s)**—usable as real data for Category 3 wind pressure channel anomaly detection; recommended surrogates: coarse-grid wind field → near-surface profile (requires multi-event CM1 data), geometry + inflow → Cp (TFCA wind tunnel archive, 63 ABL pressure files on Zenodo). The Lab does not need the two full-field D420 run directories on the cluster (~70 GB each) and will not block their cleanup

Date	What was done	One-sentence result
10-13	WRF lead reply	No ML done; only 2 events (Akron ~8 trajectories, Wister pending), insufficient to train a surrogate; a 0.01 K perturbation changes severe wind area by ~5× (internal storm spread that global models cannot resolve) → AI-generated initial conditions/ensembles not recommended; for the suggested "0.25° environmental screening," ERA5 suffices for hindcasts, and AI models add value only in forecasting settings
10-13	LBM station-by-station distance analysis	As single-station input distance to the windward face decreases from 27 → 7 m, FIR correlation increases monotonically from 0.35 → 0.67 and RMSE drops from 0.265 → 0.203; the 1h rake center column (24 m) at 0.42 matches the distance-interpolated 0.41 → supporting "distance causes correlation decay"; using all 20 rake points as input drops correlation to 0.32 (more inputs perform worse); CNN did not pass criteria at any station; x=−33 m probes lacked fluctuations (inlet boundary) and were excluded
10-13	LBM follow-up: using only true upstream probes	x≤−10 m: FIR RMSE 0.215, correlation 0.62, CNN 0.212 (1.3% better, 9/11 points); x≤−15 m: FIR 0.228, correlation 0.57, CNN 0.229. Learned models still do not clearly beat the linear transfer function; true upstream stream-wise probes at 7–12 m reach ~0.6 correlation; the digital twin wind tunnel line believes the ~0.3 correlation at the 1h rake is more likely due to probe layout differences than a sharp distance drop; station-by-station distance analysis dispatched

Date	What was done	One-sentence result
10-13	Second experiment rerun (covering 0–6 s lag, two input sets)	[Correction] The "upstream stream-wise probes" input inadvertently mixed in probes inside the cube / above the roof / in the stagnation zone (an error in the task brief discovered during review by the digital twin wind tunnel line), so FIR 0.207 / correlation 0.66 and CNN 0.220 cannot be taken as inflow predictive skill; only the 1h rake group is valid; with 1h rake inputs: FIR 0.269 vs. CNN 0.271. Learned models do not beat the linear transfer function; quasi-steady has the worst RMSE but the smallest extreme-value error (min/max Cp error 0.14/0.54 vs. FIR 0.38/0.63)
10-13	Digital twin wind tunnel lead reply	No ML done; data consists only of a single Silsoe cube case (v7, 642 s, 40 surface taps + ~70 probes, local CSV ~120 MB); only the windward face is validated; preferred surrogate: inflow time history → surface Cp(t), compared against quasi-steady theory
10-13	Second minimal experiment Round 1: LBM inflow → windward Cp(t)	TCN achieved 8% lower RMSE than linear FIR (11/11 points), but correlation was ≈0 across all three methods; lead audit: input window ≤1.3 s failed to cover the ~4 s advection lag (design flaw); Round 1 invalidated and rerun dispatched (covering 0–6 s, two input sets)
10-13 09:05	GitHub repository search (step 2 of the PI's new rule; ideas slots full, so the lead queried the GitHub API directly)	2a′ has mature general-purpose libraries (neuraloperator 3954★, Transolver 422★, MIT); 2b has Apache-2.0 unified inference frameworks (earth2studio, ai-models); thunderstorm wind 1D-CNN has no open-source implementation and must be built in-house; DeepUFNet and FoamGPT have no license
10-13	TPU Round 1 (old protocol)	5 buildings (α=1/4) × 5 wind directions, holding out T213_4: GBM lowered RMSE vs. B0 by 17% (mean) / 5% (rms), and vs. GP by only 1.5% (rms); on random wind-direction holdout, GBM lost to interpolation; 0.01 CU

Date	What was done	One-sentence result
10-13 07:18	TPU supplemented protocol	13 geometries × 2 exposures × 5 wind directions leave-one-building-out: best ML was 2.1% worse than best interpolation on mean and 0.4% better on rms, winning only 3/13 buildings each—failing the pre-specified criteria (≥20% and ≥2/3); ML was clearly better only on the boundary building T111 (1:1:1, extrapolation)
10-13 08:13	TPU all-wind-directions verification	13 geometries × 2 exposures × all official wind directions leave-one-building-out: best ML had 13% (mean) / 27% (rms) higher RMSE than the best baseline, winning only 2/13; interpolation won all 5 strictly interior buildings; on T111 extrapolation, ML still roughly halved the error (mean 0.12 vs. 0.23). ~0.06 CU, session stopped
10-13	Wind tunnel lead reply	No ML done, no measured data yet; available to borrow: YM2019 digitized 132 curves / 15,517-point CSV, synthetic data + fault-injection generator + rule-based QC checker, 148-tap table and tubing transfer function; suggested channel-by-channel anomaly detection, where a prototype can be handed back to that track to test once UF measurements arrive

Open items and next steps

- TPU result: In the interpolation regime, ML is comparable to interpolation, so this scenario is not worth investing in; **this does not mean "wind pressure surrogates are useless."** What is worth examining is extrapolation (database boundaries / new geometry types) and multi-fidelity modeling (2a′), as well as thunderstorm wind classification (3). All-wind-directions verification confirmed (10-13 08:13); waiting for the events line reply before deciding on the second experiment (≤10 CU).
- One-sentence takeaway for the PI (root lead's 10-13 wording): The interpolation scenario is not worth investing in; what is worth examining is extrapolation and multi-fidelity modeling (2a′), as well as thunderstorm wind classification (3).
- Waiting for review verdict; waiting for replies from the six lines to fill in the "Existing data" column; rerun DR once quota is restored.
- (2) After inflow correction (empty-domain r1 → v8 building case, expected from their side within this week), rerun on the new data including station-by-station distance analysis, and report the ratio of decay length (roughly estimated at ~30 m ≈ 5H in v7) to integral length scale Lu
- (3) W3 synchronized experiment: use member A′ inflow to predict member B windward 11-tap Cp(t) (same inflow, different initial fields; their side will provide small CSV paths, raw cluster data read-only), to be interpreted alongside the W3 windward difference-energy floor

- (Addendum, completed) Adding z12: rake center column FIR correlation dropped from 0.417 → 0.382 (worse), while changes at x=−10/−15/−20 were ≤0.011; CNN did not clearly beat FIR in any case, and their side withdrew the request (z12 was added by mistake in transcription; what they needed was z1/3/6, which v4b already satisfied), results archived without delivery
- Uniformly exclude the three P_x-33.05 points from all input sets (fixed boundary values, std≈0); confounder: proximity to the building cannot be separated from distance away from the inlet injection zone.
- Candidates (not dispatched): CM1 sounding → outflow statistics (waiting for ~99 cases); adding a non-learning baseline to graph-deeponet-wind; 2b GraphCast/Pangu single-event environmental comparison; 3 ASOS thunderstorm wind classification; urban wind field surrogate on the public U3DWind LBM-LES five-city dataset (card).

Literature basis

- 2a: QN6Z9F4B (PDF p.10 Fig.9), 9CBNZGZB (pp.15–17), NF6S3H9N (pp.1–3 Table 1), all in ; Taira et al. 2025.
- 2b: Feldmann et al. 2024 arXiv 2406.09474 (web search, not yet in knowledge base, pending card creation by litwiki); , .
- 3: 2T26I5GW (pp.6–8).
- 1: VNP48R7H.
- The 6 source cards listed in the task brief
- Root lead pre-search hits: ; (KAN high-rise interference effect surrogate, TPU database); (ML-augmented rectangular cylinder aerodynamic database); ,

Candidate new ideas (raised by the PI in the 10-13 discussion, not yet launched)

- AI weather foundation models for nowcasting / short-range forecasting: providing background environmental fields for tornadoes, thunderstorms, and typhoons; wind power (hub-height wind and power forecasting) may benefit more directly; reference value for CFD (inflow/boundary conditions, generative turbulence fields, methodological borrowing). (global 220 m GLES, physical model, non-AI), (generative model for urban microclimate, Liu et al. arXiv 2610.10430), (physics-constrained network for typhoon near-surface wind profiles, arXiv 2610.11405), (WeatherMesh). Approved by the PI on 10-13, sub-project opened, page; sub-project shelved on 10-13 14:00 (three-event hindcast: convective background fields only hold up to 12–24 h and are unstable, with CAPE as the bottleneck).

- 10-13, the PI: "It doesn't have to be thunderstorms—anything wind-related is fine"—see 9 brainstormed directions in , recommending A (urban / low-altitude wind forecasting chain), B (landfalling hurricane wind fields), and I (LLM-built wind damage loss vs. wind speed database), awaiting the PI's selection before opening sub-projects. On 10-13 the PI agreed to open J "Wind-engineering-specific CFD model: is the data sufficient (learning curve)", but instructed to wait until all other directions are discussed before opening them together (do not open prematurely).

- 10-14 consolidated into discussion list v2 (weather forecasting A/B/K/L/M/D/E/F, climate C/G/N, data and models J/I/H); the PI on 10-14: "CU is currently plentiful, no cap needed." Discussing item by item.

Notes from the PI

Archival addendum (10-13, root lead): events line reply on "small model for thunderstorm wind classification"

- The events line relies on Storm Events + Cook v2 Class/XClass weak labels + case-by-case radar for identification; the bottleneck is not "whether it is a thunderstorm wind," but distinguishing internal thunderstorm wind mechanisms (isolated downburst / gust front / MCS·supercell) and near-surface footprints (all 17 case-by-case reconstructions across four Texas cities were "ambiguous"). A single-station ASOS classifier offers limited help for the census; a minor potential use is finding unreported Storm Events candidates in ASOS 1-minute records. The events line is not working on this and does not plan to.
- Labels available for read-only borrowing: Cook v2 original archive (Mendeley 10.17632/88jp3swkn6.2, CC BY 4.0); Texas four-city subset (~110 MB) and ; mechanism-level labels are too scarce to train a model.
- Assessment: Recommendation for Category 3 "thunderstorm wind classification" downgraded from "Medium" to "Low" (unless narrowed to "unreported candidate screening"). Archived and shelved alongside the physical-field surrogate section; restart conditions: a need arises for large-scale unreported event screening, or sufficient mechanism-level labels become available.
- 10-14: CU is currently plentiful, so there is no need to cap CU in the first step; discuss directions one by one first, and open sub-projects together after the discussion is complete (J already approved).
- 10-14: Direction A should be subdivided—low-altitude aviation can ground flights and focuses on low wind speeds; buildings cannot be shut down and focus on high wind speeds. Building high-wind-speed prediction is relevant to active structural control as well as active wind turbine control; "this is a very good direction."
- 10-14: Under A2, both types of control (structural control and active aerodynamic shaping) can be considered, treating wind initially as a unified input; wind turbines should form a separate direction (A3), as "their community may already have some work on this."

- 10-14: A1 should also be included as a direction of interest. B (typhoons) should be the main thrust: the PI spent many years developing 3D boundary-layer models; parametric models serve wind-resistant design wind speed estimation or rapid calculation, while WRF-type downscaling serves forecasting/hindcasting (typically limited to individual typhoons); there is also the ensemble forecasting problem, which further leads into resilience and performance—planned as several sub-directions (already split into B1–B5).

- The PI, 10-14: K (AI-driven WRF for thunderstorms) and B3 (typhoons) differ fundamentally in scale and should be listed separately rather than merged; "unlimited CU" means making greater use of Colab at the current stage; the ultimate goal is at-scale forecasting and hindcasting.

- The PI, 10-14: Resilience/performance metrics cannot be validated, have low reliability, are difficult for industry to use in design, and are merely nice-to-have additions; what truly needs to be done is validation against real loss data (Direction I), which can only take initial steps for now. Climate change (G) is publishable but of modest scientific value, as there is no evidence that design wind speeds will increase.
- The PI, 10-14 correction: Do not lower the priority of B5 (resilience/performance)—research and papers can still be done, even if its practical scientific value is somewhat lower. H: Correct simulated wind fields using experiments and field measurements, starting from synoptic winds and progressing to typhoons, cyclones, and thunderstorms, with results subsequently feeding into I.

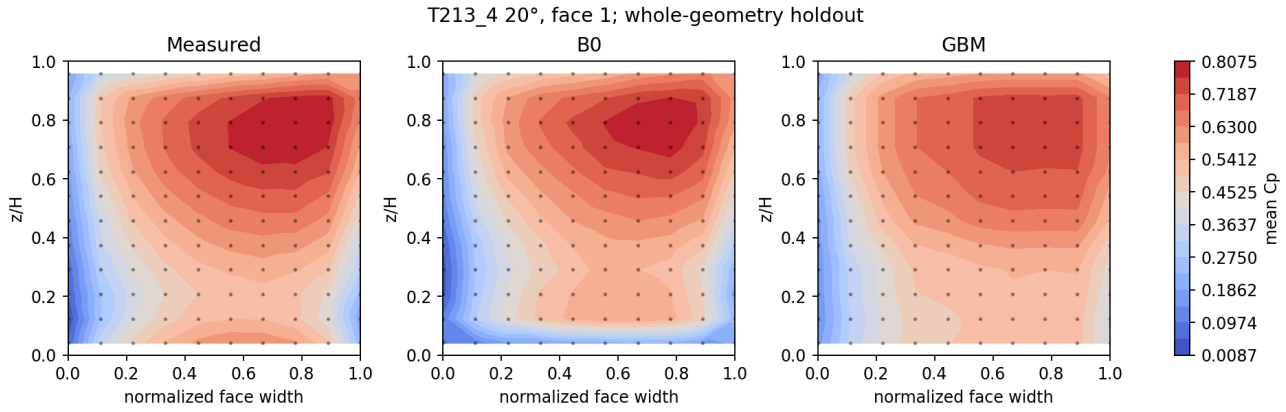
Materials from the PI (10-15): Jian Zhang, "Small language models (MiniLM) reshaping science and education"

- Materials: (paper PDF + text extracted from 100 PPT slides). Key points: uses 80–420 MB **non-generative sentence-embedding models** (all-MiniLM-L6-v2, etc.) across 550,000 PDFs from 128 geoscience journals segmented into 100 million sentences + 3 million figures for semantic retrieval; "hallucination-free" because it does not generate text, only matches original sentences with citations; proposes MiniLM+LLM ≥ LLM (i.e., RAG).
- Comparison with our current setup: ftkb already follows the same route—4,676 PDFs, 147,345 pages, FTS5 BM25 + BGE-small reranking + paraphrase-multilingual-MiniLM-L12 routing, and 105,000 vectors; the differences lie in corpus scale (~1/100), granularity (page-level vs. sentence-level), and **no figure database**.
- The lead's assessment: Small model size is not a drawback (sentence-embedding models are supposed to be small); what is worth borrowing is sentence-level corpus indexing + unsupervised clustering for literature reviews, as well as semantic retrieval over academic figures; awaiting the PI's decision on whether to open a node.
- The PI asked on 10-15: Can ftkb + the large desktop library combined rival that corpus, and can they be "added into our own large model"? The lead replied: The combined library totals ~17,000 items / 600,000 pages (vs. 550,000 papers / 100 million sentences—roughly 1/30 the document count and 1/5 the sentence count), but has a far higher density in wind engineering; "adding into the model" is actually retrieval-augmented generation, which we are already doing (pages → Claude/Gemini) without model training; our corpus was lawfully acquired by the PI personally, indexed strictly locally with page text never leaving the machine, and neither the index nor text snippets are publicly distributed.
- 10-15 inspected the author's shared MiniLM data-and-code archive (Baidu Netdisk): contains only three teaching-grade scripts for sentence-embedding retrieval, clustering, and sentiment analysis (a few dozen lines each) + a sentence-splitting shell script + 2 sample papers; **no PDF layout segmentation**

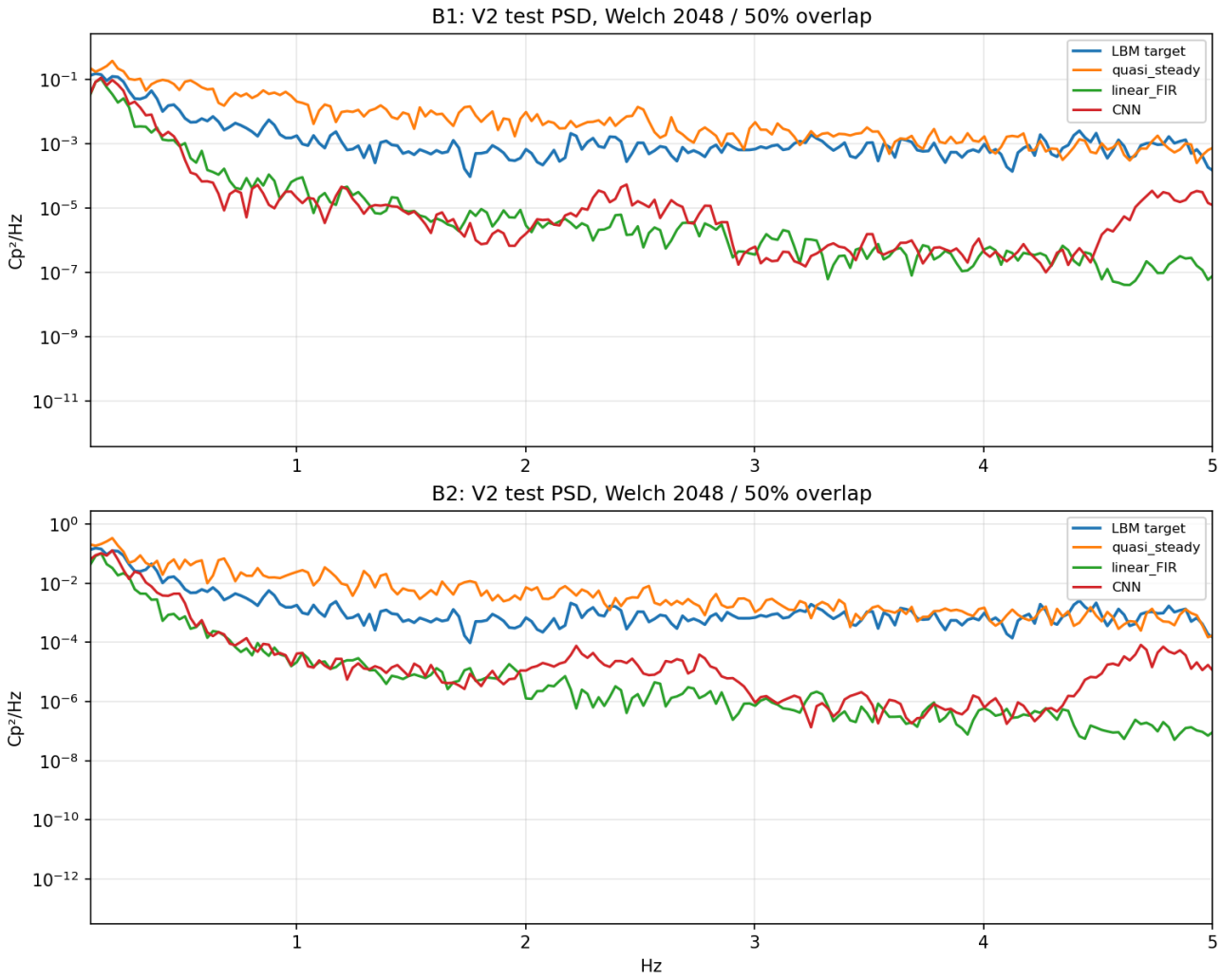
code and no figure extraction/retrieval code. Only the clustering script can be borrowed; the figure library must still be built in-house.

- 10-21 Cross-reference: The local large language model inquiry is now continuing under L32.0 (in-house teaching large language model)—the PI noted on 10-16, "This large model is very useful, we want it"; on 10-18 the L22 Strata node merged into L32.0, switching the v0 base model to Qwen3.8-Flash-Next (local, course exam with retrieval scored 9.01 vs. gpt-oss 8.54), with Stage 4 (whether to apply LoRA) pending the PI's decision. The "Literature Figure Library and Sentence-Level Retrieval" sub-project derived from these materials reached its conclusion on 10-17: pursue figure-caption retrieval (pairing 96%, figure search 19/20), and do not pursue image-to-image search or sentence-level clustering. This page remains on hold to avoid duplicating L32.0's work.
- Set by the PI on 10-21: on hold for now, not kept active, and not closed. Restart condition: When the PI raises the topic of local/in-house models again, or when L32.0's local model needs to return to research applications (batch screening, extraction, confidential data processing), resume from this page and the current state of L32.0.

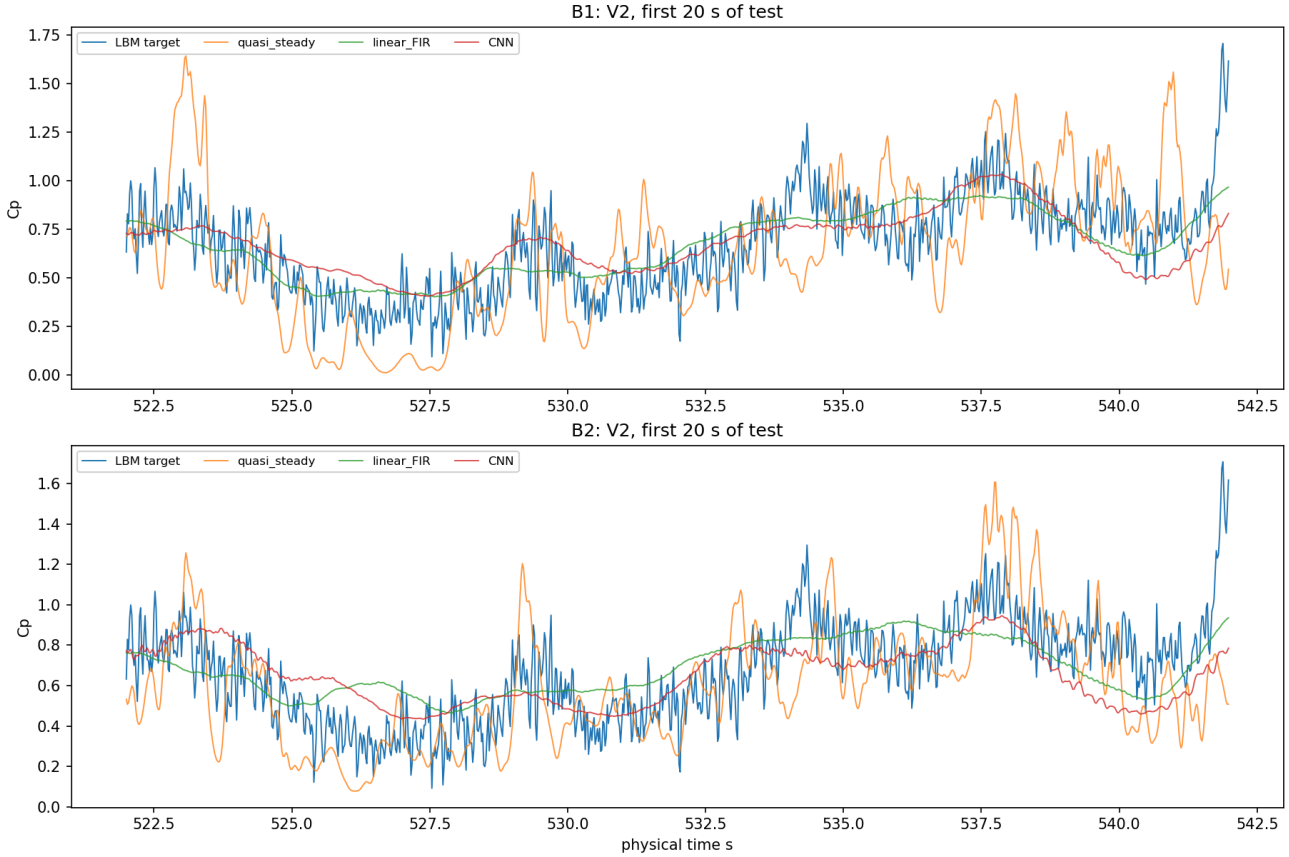
Key figures



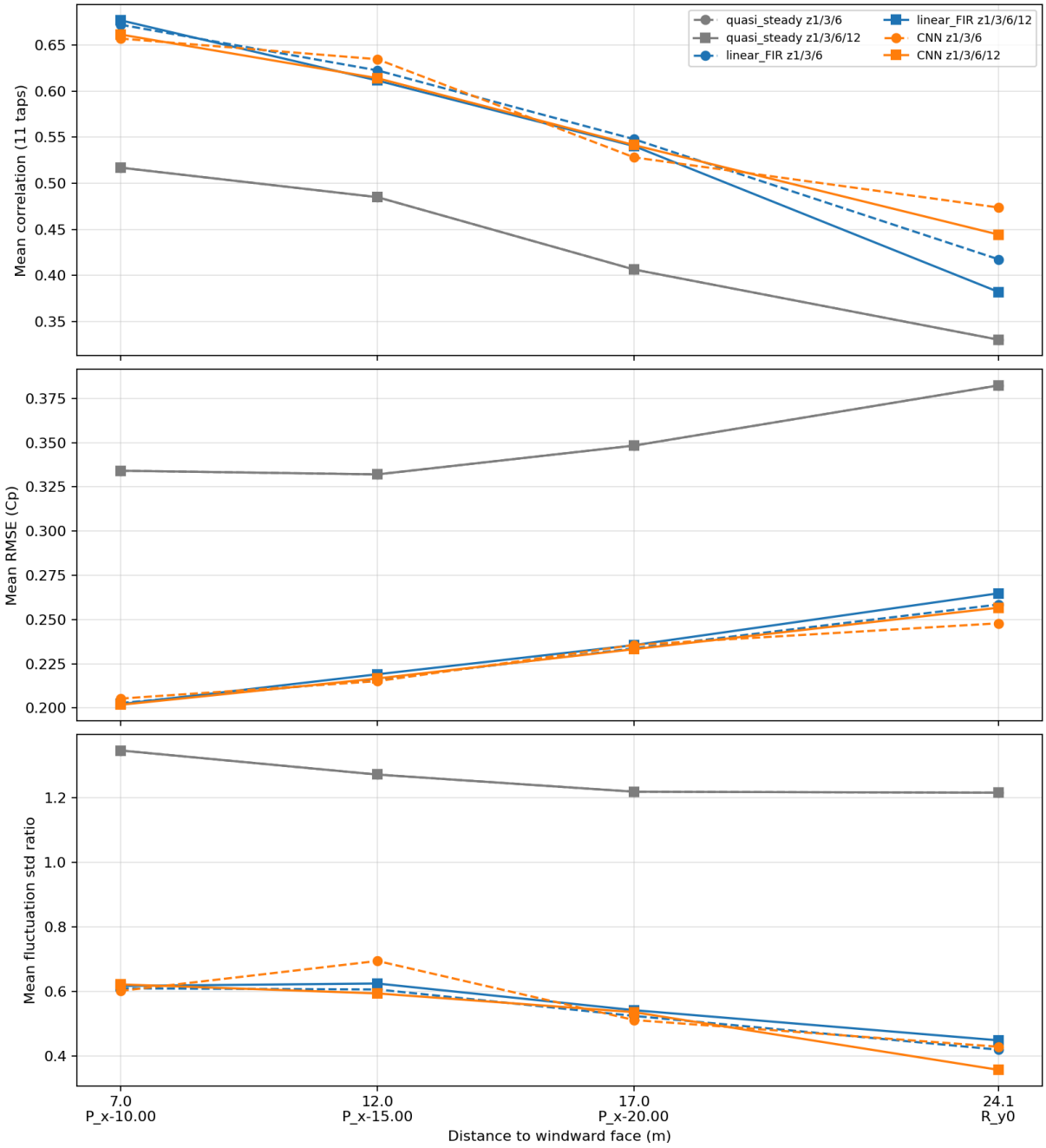
face1 mean cp r2


B1: V2 test PSD, Welch 2048 / 50% overlap
LBM target
quasi_steady
linear_FIR
CNN
Cp²/Hz
B2: V2 test PSD, Welch 2048 / 50% overlap
Hz

spectrum v3



timeseries v3



distance v4c